

COMP6237 Data Mining Introduction Lecture

Zhiwu Huang

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University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

[Book time with Zhiwu Huang: Office Hour](#)

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Module Overview

- ECS module pages [syllabus, announcements]

- <https://secure.ecs.soton.ac.uk:/module/comp6237>

COMP6237: Data Mining (2025-2026)

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Lecturer

Southampton campuses, Semester 2.

» [View notes pages](#)

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Tony Bagnall - ajb2u23@soton.ac.uk
Moderator

Lecturer tools

[Create/View Assignments](#)

When feedback has been sent to students, the lecturer should use the Handin link below and push the button on the handin page to log the date.

COMP6237 Introduction Lecture: Monday 26 Jan, 5 PM

Dear all,

(A quick heads-up and a system test.)

The COMP6237 introduction lecture is tomorrow (26 Jan), Monday 5PM, in B100 4011 (Harvard L/T B).

It should appear in your university timetable (<https://timetable.soton.ac.uk>) as well, so that's the best place to double-check.

Looking forward to seeing you there and kicking off the module together.

Best regards,

Zhiwu

[Zhiwu Huang](#) - 1 minute ago

Teaching Staff

COMP6237: Data Mining (2025-2026)

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Moderator

Southampton campuses, Semester 2.

<https://secure.ecs.soton.ac.uk:/module/comp6237>

Student Cohort

COMP6237: Data Mining (2025-2026)

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You are a lecturer on this module.

73 listed students.

[DOWNLOAD AS CSV](#) - please remember that this information is subject to data protection laws. If you are not sure what it can be used for, ask student services (SAA).

Cohorts

Show [All](#) entries

Search:

Count	Year	Deg.	Degree	Prog.	Programme
1	05	6190	MEng Software Engineering with Industrial Studies		
1	02	4475	MSc Artificial Intelligence		
2	04	4439	Artificial Intelligence (MEng Electronic Engineering)		
3	04	4444	Artificial Intelligence (MEng Computer Science)		
4	01	4466	MSc Computer Science		
7	04	4443	Computer Science (MEng Computer Science)		
20	01	4475	MSc Artificial Intelligence		
35	01	6150	MSc Data Science		

<https://secure.ecs.soton.ac.uk:/module/comp6237>

Module Overview

- **Course website [notes, slides, recordings]**
 - <https://comp6237.ecs.soton.ac.uk/zh.html>



COMP6237 Data Mining
Notes, Slides and Demos for COMP6237 2025-26

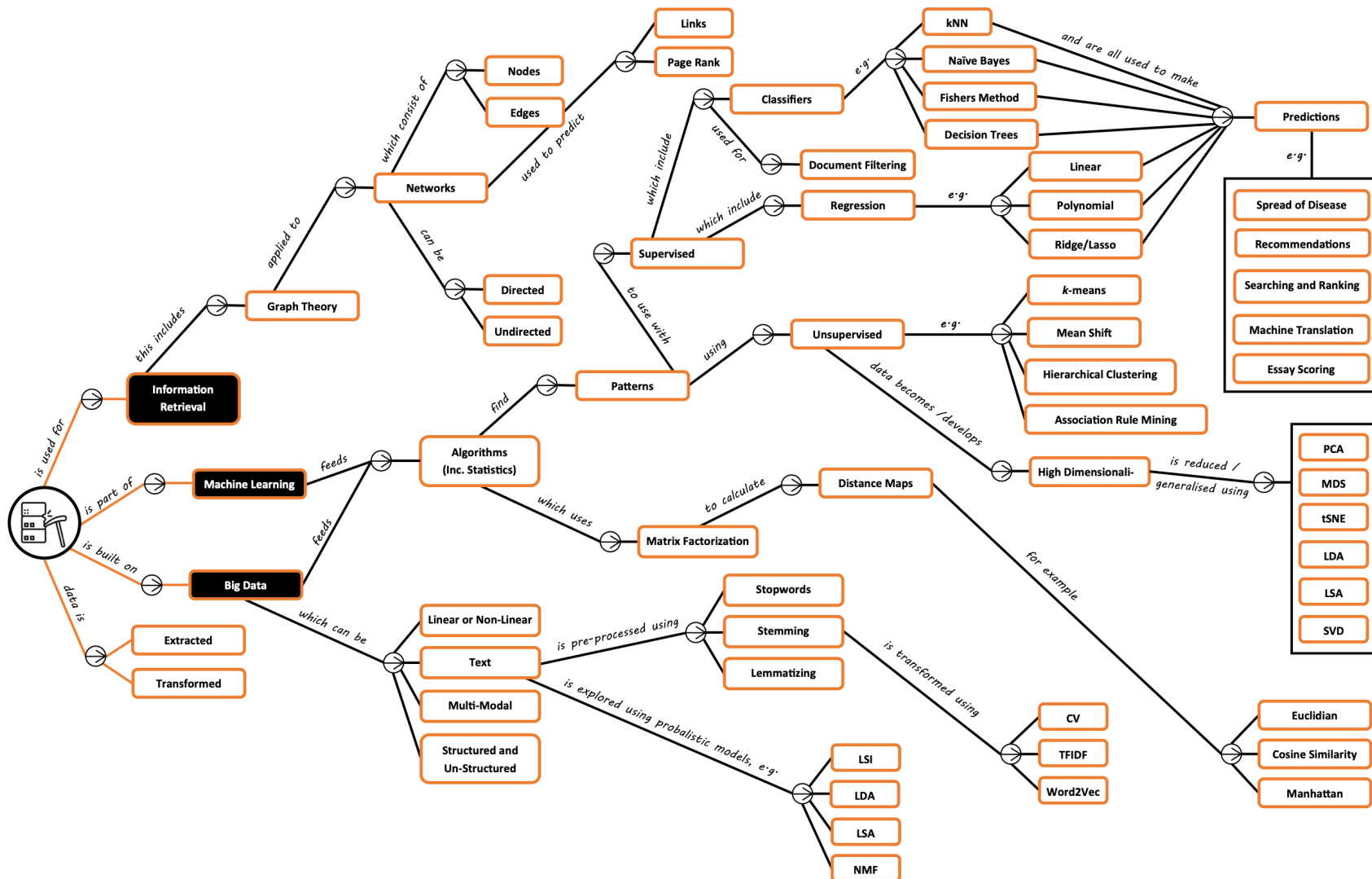
Maintained by **Dr Shoaib Ehsan** and **Dr Zhiwu Huang**

Date	Title	Slides	Handouts	Code	Video
26/01/26	Introduction to Data Mining	PDF	–	–	–
29/01/26	Discovering Groups	PDF	–	git	–
30/01/26	Covariance, EVD, PCA & SVD	PDF	–	git	–

Module Overview

- Developed by Prof Jon Hare & Dr Jo Grundy, run for the 9th time
- Created to fill a gap
 - **Data mining is almost synonymous with machine learning**
 - Inevitably have some overlap with machine learning modules
e.g. COMP3222/COMP3223/COMP6245/COMP6208
 - Should be complementary and offer different views
 - **Slightly more applied pragmatic focus**
 - How do you work with real world data?
 - How do you solve real problems?

Module Overview



- Around 26 lectures + additional tutorials
- Wide range of data mining topics

Module Overview

- Reading material

- Toby Segaran. Programming Collective Intelligence: Building Smart Web 2.0 Applications. O'Reilly, 2007
- Aurélien Géron. Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. O'Reilly Media. March 2017
- J. Leskovec et al. Mining of Massive Datasets. Third Edition. Cambridge University Press. 2020
- M. J. Zaki and W. Meira, Data Mining and Machine Learning: Fundamental Concepts and Algorithms. Cambridge University Press. 2020.

Module Timetable

Day	Time	Room
Monday	5 PM	B100 4011 (Harvard L/T B)
Tuesday	5 PM	B46 2003 (L/T B)
Thursday	12 PM	B46 2003 (L/T B)
Friday	3 PM	B46 2003 (L/T B)

Date	Semester Week	Lecturer(s)	Topic/Title
26-Jan	1	Zhiwu	Intro to Data Mining
29-Jan		Zhiwu	Finding Groups
30-Jan		Zhiwu	Covariance
02-Feb	2	Zhiwu	Embedding Data
03-Feb		Zhiwu	Search
06-Feb		Shoaib	Linear Regression I; Group CW set
09-Feb	3	Shoaib	Linear Regression II
12-Feb		Shoaib	Linear Regression Problem Sets

<http://comp6237.ecs.soton.ac.uk/>

Note: This may sometimes also change –we'll update you by email (check ECS module page)

Module Assessment

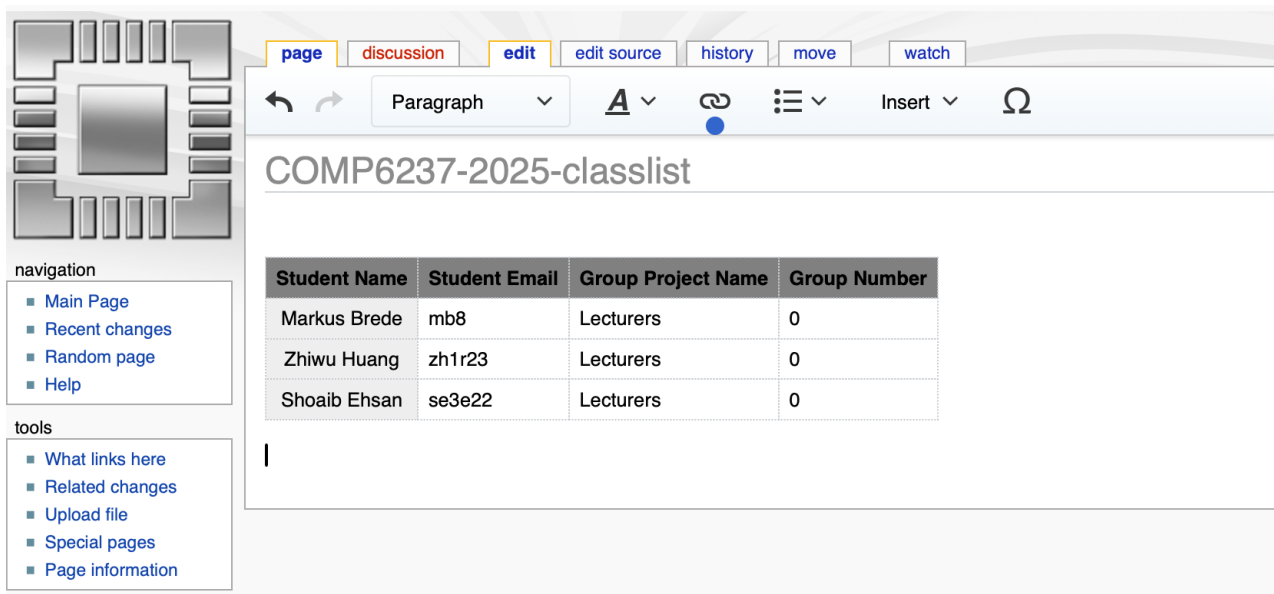
Assessment

Summative

Assessment method	Contribution to final mark
Group Coursework	30%
Final Exam	70%

Group Coursework

- Three Q & A sessions in **Week 3&4**; by that time we want you to have formed groups at <https://secure.ecs.soton.ac.uk/student/wiki/w/COMP6237-2025-classlist>



The screenshot shows a MediaWiki page titled "COMP6237-2025-classlist". The page has a navigation sidebar on the left with links like "Main Page", "Recent changes", "Random page", and "Help". The main content area contains a table with student information. The table has four columns: "Student Name", "Student Email", "Group Project Name", and "Group Number". The table lists three students: Markus Brede, Zhiwu Huang, and Shoaib Ehsan, all of whom are in the "Lecturers" group with a group number of 0. The page also features a top navigation bar with tabs for "page", "discussion", "edit", "edit source", "history", "move", and "watch".

Student Name	Student Email	Group Project Name	Group Number
Markus Brede	mb8	Lecturers	0
Zhiwu Huang	zh1r23	Lecturers	0
Shoaib Ehsan	se3e22	Lecturers	0

- Four presentation sessions before Easter (**Week 8**)
- Report submission at the end of the term (**May 15**)

Final Exam

- **Computer-aided with only multiple-choice questions**
 - Shoaib (40 marks) + Zhiwu (40 Marks) + Markus (20 Marks)
 - 3 Lectures for revisions
 - Platform: <https://moodle.ecs.soton.ac.uk>

Please read the instructions below and wait on this page until an invigilator tells you to start. Press "Start Exam" when instructed to by an invigilator.

This is a Computer Aided Assessment. Follow all instructions in the exam software.

SEMESTER 2 EXAMINATIONS 2024-2025

Data Mining

Duration: 120 mins (2 hours)

This paper contains 33 questions.

Answer all questions.

Only University approved calculators may be used.

A foreign language dictionary is permitted ONLY IF it is a paper version of a direct 'Word to Word' translation dictionary AND it contains no notes, additions or annotations.

What is Data Mining?

“Data mining is **the process of extracting and finding patterns in massive data sets** involving methods at the intersection of machine learning, statistics, and database systems.

Data mining is an interdisciplinary subfield of computer science and statistics with an overall goal of **extracting information (with intelligent methods) from a data set and transforming the information into a comprehensible structure for further use.**”

–Wikipedia

Why Do we Learn Data Mining?

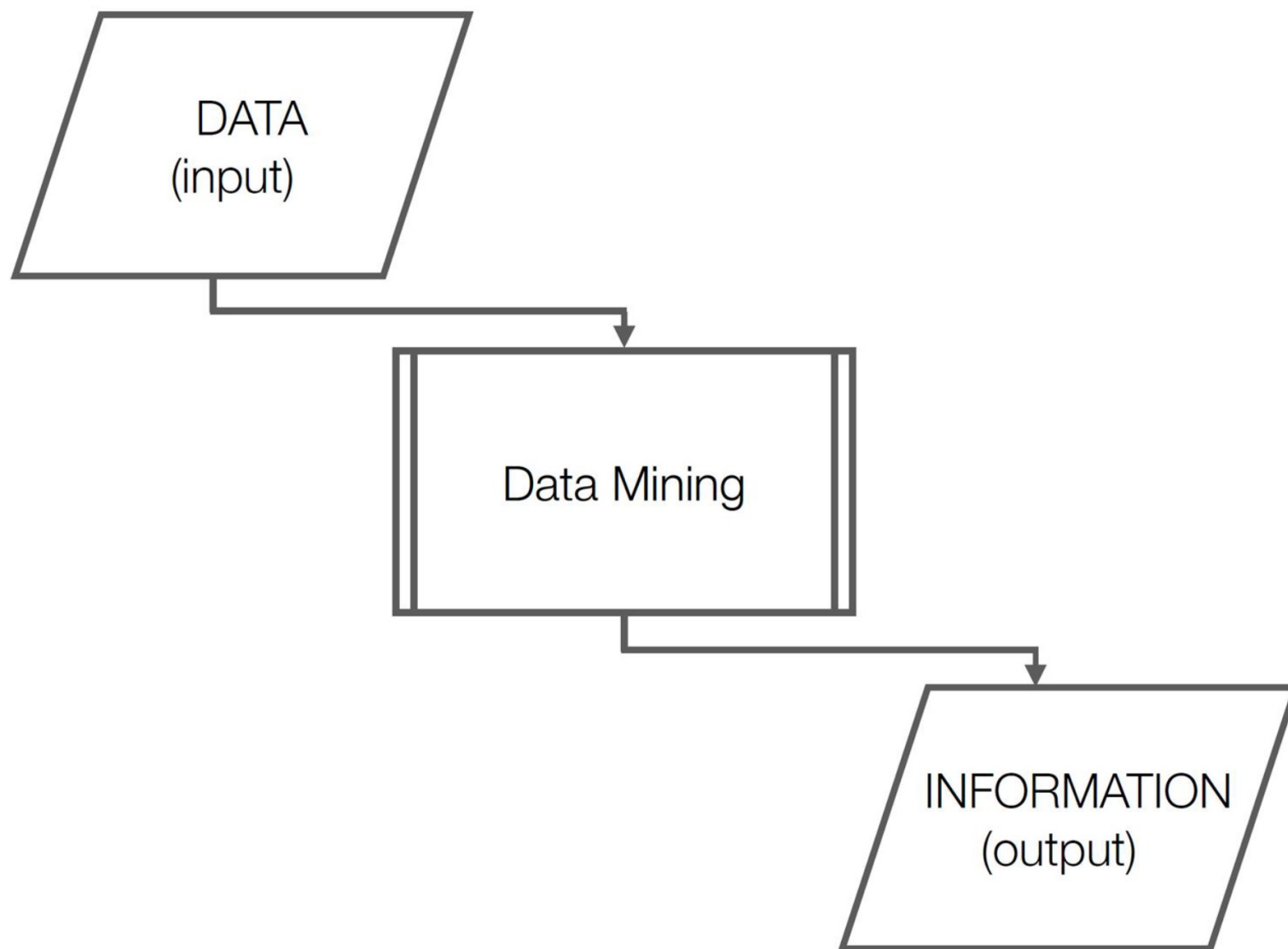
- “Modern AI systems like ChatGPT are trained on massive amounts of data and can understand text, images, and tables. So why do we still need to learn data mining?”

Why Do we Learn Data Mining?

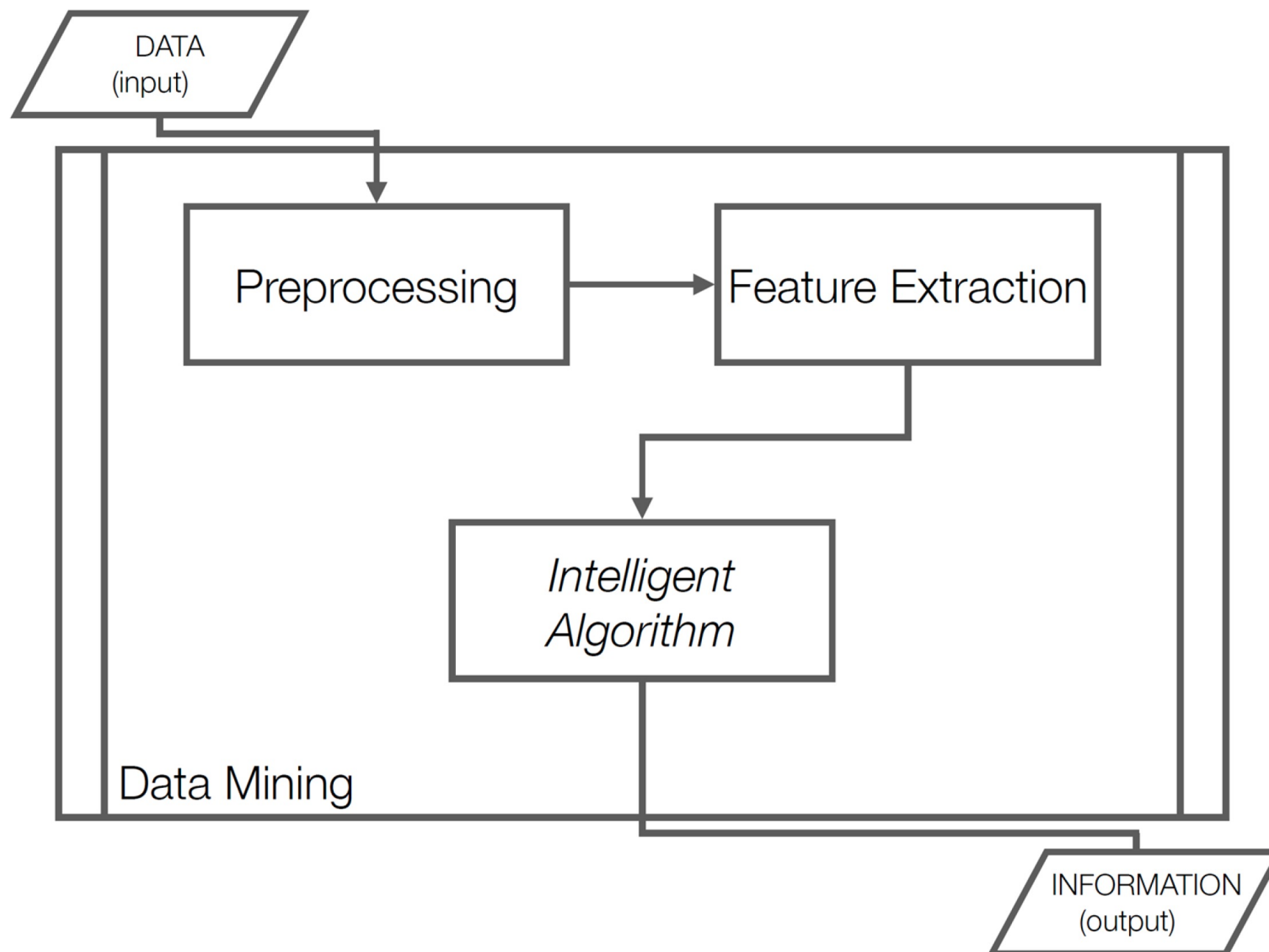
- You might ask ChatGPT: “Hi ChatGPT, this is my company’s customer data. Please use data-mining techniques to discover meaningful patterns, customer segments, and insights that are not immediately obvious from the raw data.”

Task	Data Mining Skills (Student)	AI Capability	Why Students Still Need Data Mining
Problem definition	Turns real-world needs into data questions	Responds to given prompts	AI can’t choose the right problem
Data preparation	Cleans data, fixes errors, handles bias	Suggests generic steps	Bad data leads to wrong results
Pattern discovery	Evaluates significance and stability	Finds correlations	AI may find misleading patterns
Model evaluation	Selects proper metrics and validation	Computes metrics	Wrong metrics = wrong decisions
Interpretation	Links results to domain actions	Summarizes outputs	AI lacks real-world accountability

How Can We Do Data Mining?



How Can We Do Data Mining?



Descriptive Techniques

PCA

ICA

MDS

Clustering

Anomaly Detection

...

*Intelligent
Algorithm*

Predictive Techniques

Classification

Ranking

Regression

Matrix Completion

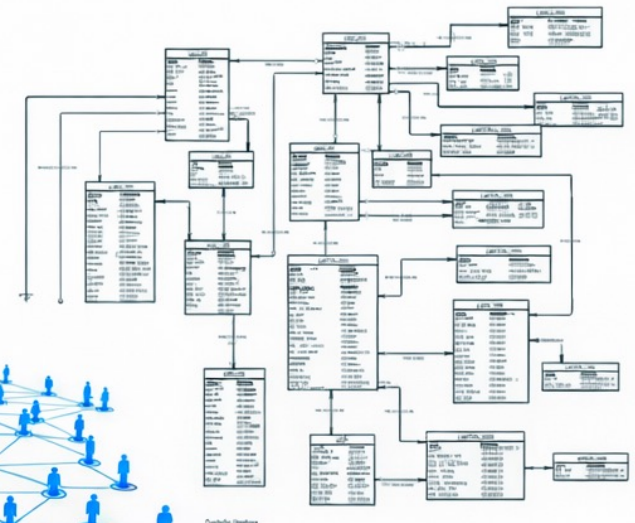
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The screenshot shows the Microsoft Excel application window with a spreadsheet titled "Sales". The spreadsheet contains the following data:

Date	Task	Area	Amount
1/1/2018	Task 1	Area 1	\$1,000.00
1/1/2018	Task 1	Area 2	\$1,000.00
1/1/2018	Task 1	Area 3	\$1,000.00
1/1/2018	Task 1	Area 4	\$1,000.00
1/1/2018	Task 1	Area 5	\$1,000.00
1/1/2018	Task 1	Area 6	\$1,000.00
1/1/2018	Task 1	Area 7	\$1,000.00
1/1/2018	Task 1	Area 8	\$1,000.00
1/1/2018	Task 1	Area 9	\$1,000.00
1/1/2018	Task 1	Area 10	\$1,000.00
1/1/2018	Task 1	Area 11	\$1,000.00
1/1/2018	Task 1	Area 12	\$1,000.00
1/1/2018	Task 1	Area 13	\$1,000.00
1/1/2018	Task 1	Area 14	\$1,000.00
1/1/2018	Task 1	Area 15	\$1,000.00
1/1/2018	Task 1	Area 16	\$1,000.00
1/1/2018	Task 1	Area 17	\$1,000.00
1/1/2018	Task 1	Area 18	\$1,000.00
1/1/2018	Task 1	Area 19	\$1,000.00
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1/1/2018	Task 1	Area 25	\$1,000.00
1/1/2018	Task 1	Area 26	\$1,000.00
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1/1/2018	Task 1	Area 31	\$1,000.00
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1/1/2018	Task 1	Area 33	\$1,000.00
1/1/2018	Task 1	Area 34	\$1,000.00
1/1/2018	Task 1	Area 35	\$1,000.00
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1/1/2018	Task 1	Area 41	\$1,000.00
1/1/2018	Task 1	Area 42	\$1,000.00
1/1/2018	Task 1	Area 43	\$1,000.00
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1/1/2018	Task 1	Area 48	\$1,000.00
1/1/2018	Task 1	Area 49	\$1,000.00
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1/1/2018	Task 1	Area 51	\$1,000.00
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1/1/2018	Task 1	Area 68	\$1,000.00
1/1/2018	Task 1	Area 69	\$1,000.00
1/1/2018	Task 1	Area 70	\$1,000.00
1/1/2018	Task 1	Area 71	\$1,000.00
1/1/2018	Task 1	Area 72	\$1,000.00
1/1/2018	Task 1	Area 73	\$1,000.00



The Plan for the Next 12 Weeks

.You will learn to solve real-world problems – e.g.:

- Recommender systems
- Market Basket Analysis
- Document filtering and spam detection
- Duplicate document detection
- Link prediction
- Community detection
- Ranking search results
- Social network analysis

.You will also learn various tools & techniques - e.g.:

- Linear algebra (SVD, Eigendecomposition & PCA, NNMF, etc.)
- Optimisation (e.g. stochastic gradient descent)
- Dynamic programming (frequent itemsets)
- Hashing (LSH, Sketching, Bloom Filters)
- Statistics of regression analysis
- Information theory
- Network theory

The Group Coursework

- .You need to form groups
 - Target size is 4 (**strictly**)
 - As a group, you need to choose a data mining problem to work on
 - (You'll need to train and evaluate models and compare their performance [possibly against approaches from others])
- . Come along to the slots in week 3 to discuss your ideas for problems to work on with us
- .Enter your team name and team members on the student wiki:

<https://secure.ecs.soton.ac.uk/student/wiki/w/COMP6237-2025-classlist>

Key Date

- Each team needs to submit a 1-page project brief by the end of week 4 (**20th of Feb**).
- Before Easter groups must present their idea and approaches to the class.
 - ❖ Teams should be prepared to present in the first slot; to ensure fairness we will pick teams at random
- Teams must submit a conference paper by **4pm on May 15**.